

MGTECON 603 - Problem Set 6

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0 Summary Statistics

The summary statistics of the data are shown in table A.1 in the appendix. Also, the scatterplot and the correlation matrix of the data are shown in figure A.1 in the appendix.

1 Problem 1

1.1 Part (a)-(c)

I estimate the OLS model with the following formula using two different standard error estimators: homoskedastic and cluster-robust.

$$Y_i = \alpha + \beta X_i + \epsilon_i \tag{1}$$

The results are shown in table 1, where the first column(1) is the homoskedastic standard error estimator and the second column(2) is the cluster-robust standard error estimator.

The OLS standard errors for (α, β) are (0.0045, 0.0003) and (0.0307, 0.0018) for the homoskedastic and cluster-robust standard error(CRSE) estimators, respectively.

Dependent Variable:	lwage	
Model:	(1)	(2)
<i>Variables</i>		
Constant	4.995*** (0.0045)	4.995*** (0.0307)
educ	0.0708*** (0.0003)	0.0708*** (0.0018)
<i>Fit statistics</i>		
Observations	329,509	329,509
R ²	0.11729	0.11729
Adjusted R ²	0.11729	0.11729
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table 1: Regression of lwage on educ

1.2 Part (d)

I calculate the homoskedastic and cluster-robust standard errors by bootstrapping. In the case of homoskedastic OLS, I used the parametric bootstrap method, since it directly assumes the homoskedasticity of the error term. I followed the following algorithms:

Algorithm 1 Parametric Bootstrap for OLS Standard Errors

- 1: Estimate the initial OLS model $lwage_i = \beta_0 + \beta_1 educ_i + \epsilon_i$ to obtain coefficients $\hat{\beta}_0, \hat{\beta}_1$ and residuals $\{\hat{\epsilon}_i\}_{i=1}^n$.
 - 2: **for** $b = 1$ to B **do**
 - 3: Create a bootstrap sample of residuals $\{\epsilon_i^{*b}\}_{i=1}^n$ by sampling with replacement from the original residuals $\{\hat{\epsilon}_i\}_{i=1}^n$.
 - 4: Generate a new outcome variable $lwage_i^{*b} = \hat{\beta}_0 + \hat{\beta}_1 educ_i + \epsilon_i^{*b}$ for $i = 1, \dots, n$.
 - 5: Re-estimate the OLS model of $lwage^{*b}$ on $educ$ to get the coefficient estimate $\hat{\beta}_1^{*b}$.
 - 6: Store $\hat{\beta}_1^{*b}$.
 - 7: **end for**
 - 8: The bootstrapped standard error is the standard deviation of the stored coefficients:

$$SE(\hat{\beta}_1) = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\beta}_1^{*b} - \bar{\beta}_1^*)^2}.$$
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For cluster-robust standard errors, I use a two-stage cluster bootstrap:

Algorithm 2 Cluster-Robust Bootstrap (Two-Stage Resampling)

- 1: Identify the M unique clusters (states of birth) in the data.
 - 2: **for** $b = 1$ to B **do**
 - 3: Create a bootstrap sample of M cluster IDs, $\{c_1^{*b}, \dots, c_M^{*b}\}$, by sampling with replacement from the original cluster IDs.
 - 4: Initialize an empty bootstrap dataset, **data_b**.
 - 5: **for** each sampled cluster ID c_j^{*b} in the list **do**
 - 6: Let the original data for this cluster be **data_c**.
 - 7: Create a bootstrap sample of individuals from this cluster by sampling $|\mathbf{data_c}|$ rows with replacement from **data_c**.
 - 8: Append these resampled individuals to **data_b**.
 - 9: **end for**
 - 10: Re-estimate the OLS model of *lwage* on *educ* using the bootstrap dataset **data_b** to get the coefficient estimate $\hat{\beta}_1^{*b}$.
 - 11: Store $\hat{\beta}_1^{*b}$.
 - 12: **end for**
 - 13: The cluster-robust bootstrapped standard error is the standard deviation of the stored coefficients: $SE_{cluster}(\hat{\beta}_1) = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\beta}_1^{*b} - \bar{\beta}_1^*)^2}$.
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The results are shown in table 2.

OLS Standard Error	Cluster-Robust Standard Error
0.0003	0.0018

Table 2: Comparison of OLS and Cluster-Robust Standard Errors

1.3 Part (e)-(g)

I estimate the OLS model with the following formula:

$$Y_i = \alpha + \beta_1 X_i + \beta_2 \bar{X}_c + \epsilon_i \quad (2)$$

The results are shown in table 3, where the first column(1) is the OLS standard error estimator and the second column(2) is the cluster-robust standard error estimator.

Dependent Variable: Model:	lwage	
	(1)	(2)
<i>Variables</i>		
Constant	4.154*** (0.0186)	4.154*** (0.1579)
educ	0.0671*** (0.0003)	0.0671*** (0.0014)
mean_educ	0.0696*** (0.0015)	0.0696*** (0.0120)
<i>Fit statistics</i>		
Observations	329,509	329,509
R ²	0.12305	0.12305
Adjusted R ²	0.12305	0.12305
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table 3: Regression of lwage on mean educ

The OLS standard errors for $(\alpha, \beta_1, \beta_2)$ are (0.0186, 0.0003, 0.0015) and (0.1579, 0.0014, 0.0120) for the homoskedastic and cluster-robust standard error(CRSE) estimators, respectively.

1.4 Part (h)

I calculate the homoskedastic and cluster-robust standard errors by bootstrapping. In the case of homoskedastic OLS, I used the parametric bootstrap method, since it directly assumes the homoskedasticity of the error term. I followed the same algorithm as in part (d); the only difference is that I now have two coefficients to estimate: β_1 and β_2 .

The results are shown in table 4.

Estimator	SE(educ)	SE(mean_educ)
OLS	0.0004	0.0015
CRSE	0.0014	0.0123

Table 4: Comparison of Bootstrapped OLS and Cluster-Robust Standard Errors

1.5 Summary of Results

The results from the previous sections are summarized in tables 5 and 6.

Table 5: Summary of Standard Errors for Model 1: $lwage \sim educ$

Standard Error Type	Analytic	Bootstrap
Homoskedastic (OLS)	0.0003	0.0003
Cluster-Robust (CRSE)	0.0018	0.0018

Notes: The coefficient for ‘educ’ is 0.0708. The analytic SEs are from the ‘fixest::feols’ output. The bootstrap SEs are the standard deviation of the bootstrap distribution from 10,000 replications.

Table 6: Summary of Standard Errors for Model 2: $lwage \sim educ + mean_educ$

Coefficient	Analytic		Bootstrap	
	Homoskedastic	Cluster-Robust	Homoskedastic	Cluster-Robust
educ (0.0671)	0.0003	0.0014	0.0004	0.0014
mean_educ (0.0696)	0.0015	0.0120	0.0015	0.0123

Notes: The analytic SEs are from the ‘fixest::feols’ output. The bootstrap SEs are the standard deviation of the bootstrap distribution from 10,000 replications.

2 Problem 2

Let the model be

$$Y_{ic} = \alpha + \beta' \mathbf{X}_{ic} + \eta_c + \nu_{ic} \quad (3)$$

where the composite error term is

$$v_{ic} = \eta_c + \nu_{ic} \quad (4)$$

The full variance-covariance matrix of the errors is block-diagonal, where $\mathbf{I}_C$ is the identity matrix of size C and $\mathbf{\Omega}_c$ is the covariance matrix for a single cluster c (size N_c):

$$\mathbf{\Omega} = \mathbf{I}_C \otimes \mathbf{\Omega}_c \quad (5)$$

We can write the covariance matrix for a single cluster c as:

$$\mathbf{\Omega}_c = \text{Var}(\mathbf{v}_c) = \sigma_\nu^2 \mathbf{I}_{N_c} + \sigma_\eta^2 \mathbf{J}_{N_c} \quad (6)$$

The GLS estimator can be written as follows:

$$\hat{\beta}_{GLS} = \arg \min_{\beta} (\mathbf{Y} - \mathbf{X}\beta)' \mathbf{\Omega}^{-1} (\mathbf{Y} - \mathbf{X}\beta) \quad (7)$$

$$= (\mathbf{X}' \mathbf{\Omega}^{-1} \mathbf{X})^{-1} (\mathbf{X}' \mathbf{\Omega}^{-1} \mathbf{Y}) \quad (8)$$

Then, we can think of the GLS estimator as the OLS estimator of the transformed variables:

$$\mathbf{X}^* = \mathbf{P}\mathbf{X} = \mathbf{\Omega}^{-1/2} \mathbf{X} \quad (9)$$

The transformation for observations within cluster c is equivalent to some sort of de-meaning process:

$$\mathbf{X}_{ic}^* = \mathbf{X}_{ic} - \lambda_c \bar{\mathbf{X}}_c \quad (10)$$

where $\bar{\mathbf{X}}_c$ is the cluster mean of $\mathbf{X}$, and λ_c is the quasi-demeaning factor:

$$\lambda_c = 1 - \frac{\sigma_\nu}{\sqrt{\sigma_\nu^2 + N_c \sigma_\eta^2}} \quad (11)$$

Since the regressor is constant within the cluster, we have:

$$\mathbf{X}_{ic} = \mathbf{X}_c \quad \text{for all } i \quad (12)$$

This implies that the cluster mean of the regressor is simply the regressor itself:

$$\bar{\mathbf{X}}_c = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{X}_{ic} = \mathbf{X}_c \quad (13)$$

Substituting this into the transformation for $\mathbf{X}_{ic}^*$:

$$\mathbf{X}_{ic}^* = \mathbf{X}_c - \lambda_c \mathbf{X}_c \quad (14)$$

$$\mathbf{X}_{ic}^* = (1 - \lambda_c) \mathbf{X}_c \quad (15)$$

Since $\mathbf{X}_{ic}^*$ is proportional to $\mathbf{X}_c$, the transformed regressor matrix $\mathbf{X}^*$ is proportional to the original regressor matrix $\mathbf{X}$:

$$\mathbf{X}^* = (1 - \lambda) \mathbf{X} \quad (16)$$

where λ is constant across clusters (since N_c is constant across clusters, $N_c = N/C$). The GLS estimator is the OLS estimator of the transformed model $\mathbf{Y}^* = \mathbf{X}^*\boldsymbol{\beta} + \mathbf{v}^*$:

$$\hat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}^{*'}\mathbf{X}^*)^{-1}(\mathbf{X}^{*'}\mathbf{Y}^*) \quad (17)$$

Substitute the proportional relationships ($\mathbf{X}^* = (1 - \lambda)\mathbf{X}$ and $\mathbf{Y}^* = (1 - \lambda)\mathbf{Y}$):

$$\hat{\boldsymbol{\beta}}_{GLS} = (((1 - \lambda)\mathbf{X})'((1 - \lambda)\mathbf{X}))^{-1}(((1 - \lambda)\mathbf{X})'((1 - \lambda)\mathbf{Y})) \quad (18)$$

$$= ((1 - \lambda)^2\mathbf{X}'\mathbf{X})^{-1}((1 - \lambda)^2\mathbf{X}'\mathbf{Y}) \quad (19)$$

$$= \frac{1}{(1 - \lambda)^2}(\mathbf{X}'\mathbf{X})^{-1}(1 - \lambda)^2(\mathbf{X}'\mathbf{Y}) \quad (20)$$

$$= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{Y}) = \hat{\boldsymbol{\beta}}_{OLS} \quad (21)$$

The estimators are algebraically identical

$$\hat{\boldsymbol{\beta}}_{GLS} = \hat{\boldsymbol{\beta}}_{OLS} \quad (22)$$

which implies that their variances are also equal:

$$\text{Var}(\hat{\boldsymbol{\beta}}_{GLS}) = \text{Var}(\hat{\boldsymbol{\beta}}_{OLS}) \quad (23)$$

A Summary Statistics

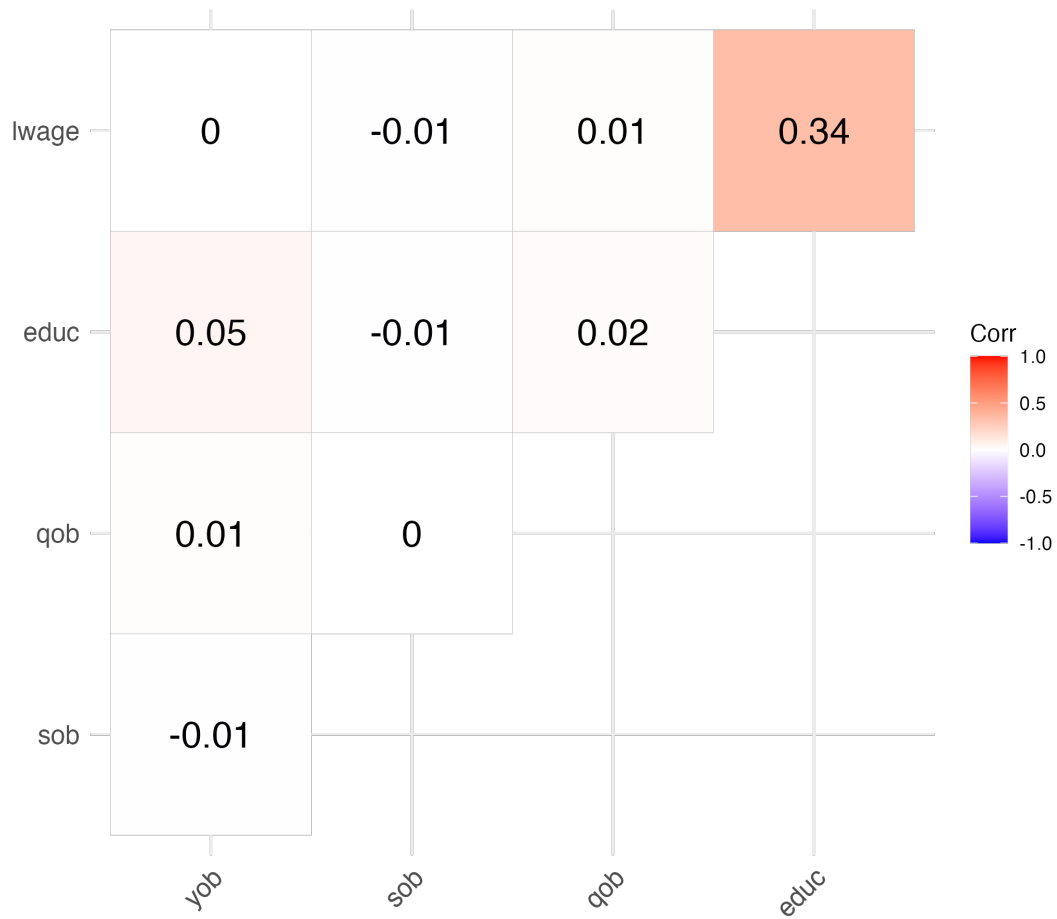


Figure A.1: Correlation Matrix Plot

variable	numNA	numZeros	fracZeros	mean	sd	min	max	10%	20%	30%	40%	50%	90%	95%	99%	99.9%
yob	0	0	0.0000	34.6028	2.9050	30.0000	39.0000	30.0000	32.0000	33.0000	34.0000	35.0000	39.0000	39.0000	39.000	39.0000
sob	0	0	0.0000	30.6926	14.2184	1.0000	56.0000	9.0000	17.0000	22.0000	26.0000	34.0000	48.0000	53.0000	55.000	56.0000
qob	0	0	0.0000	2.5064	1.1119	1.0000	4.0000	1.0000	1.0000	2.0000	2.0000	3.0000	4.0000	4.0000	4.000	4.0000
educ	0	598	0.0018	12.7699	3.2812	0.0000	20.0000	9.0000	11.0000	12.0000	12.0000	12.0000	17.0000	19.0000	20.000	20.0000
lwage	0	3	0.0000	5.8999	0.6788	-2.3418	10.5321	5.2343	5.5218	5.7243	5.8472	5.9525	6.5567	6.8004	7.274	7.9495

Table A.1: Summary Statistics